# MSG No. 19.30 $P_I2_12_12_1$ [ Type IV, orthorhombic ]

Table 1: Wyckoff site: `4a`, site symmetry: `2'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\, 0,\, \frac{1}{4}\right]$ | `[1,6]` |
| 2 | $\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{3}{4}\right]$ | `[2,5]` |
| 3 | $\left[-x,\, \frac{1}{2},\, \frac{1}{4}\right]$ | `[3,8]` |
| 4 | $\left[\frac{1}{2}-x,\, 0,\, \frac{3}{4}\right]$ | `[4,7]` |

Table 2: Wyckoff site: `4b`, site symmetry: `.2'.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\, y,\, 0\right]$ | `[1,7]` |
| 2 | $\left[\frac{3}{4},\, \frac{1}{2}-y,\, 0\right]$ | `[2,8]` |
| 3 | $\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$ | `[3,5]` |
| 4 | $\left[\frac{1}{4},\, -y,\, \frac{1}{2}\right]$ | `[4,6]` |

Table 3: Wyckoff site: `4c`, site symmetry: `..2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\, \frac{1}{4},\, z\right]$ | `[1,8]` |
| 2 | $\left[\frac{1}{2},\, \frac{1}{4},\, -z\right]$ | `[2,7]` |
| 3 | $\left[0,\, \frac{3}{4},\, \frac{1}{2}-z\right]$ | `[3,6]` |
| 4 | $\left[\frac{1}{2},\, \frac{3}{4},\, z+\frac{1}{2}\right]$ | `[4,5]` |

Table 4: Wyckoff site: `8d`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\, y,\, z]$ | `[1]` |
| 2 | $\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$ | `[2]` |
| 3 | $\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$ | `[3]` |
| 4 | $\left[\frac{1}{2}-x,\, -y,\, z+\frac{1}{2}\right]$ | `[4]` |
| 5 | $\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$ | `[5]` |
| 6 | $\left[x,\, -y,\, \frac{1}{2}-z\right]$ | `[6]` |
| 7 | $\left[\frac{1}{2}-x,\, y,\, -z\right]$ | `[7]` |
| 8 | $\left[-x,\, \frac{1}{2}-y,\, z\right]$ | `[8]` |